

Saiyed Saizan Shahnawaz

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Education

University of Wisconsin–Madison

Expected May 2028

B.S. Computer Science GPA: 3.77/4.0

Madison, WI

Relevant Coursework: Data Structures and Algorithms, Discrete Mathematics, Systems Programming, Object-Oriented Programming, Linear Algebra.

Technical Skills

Programming: TypeScript, Python, JavaScript, Java, SQL, C, HTML/CSS

Frameworks and Systems: FastAPI, Hono, Node.js, REST APIs, PostgreSQL, SQLite, Firestore, Chrome Extension APIs, Docker, Linux, Git, pytest, Vitest

AI/ML: LLM tool-calling, agent orchestration, retrieval pipelines, structured outputs, self-hosted inference, LeRobot, Isaac Sim

Experience

Flow – Privacy-First AI Campus Assistant

2026

Summer Research Intern, UW–Madison Summer AI Lab (team of 3); 3rd place, Entra AI Club Hackathon Madison, WI

- Built a DARS degree-audit parser extracting completed courses, requirements, and remaining needs from student-uploaded PDFs via deterministic rule- and regex-based extraction over **pdfplumber**, with a 180-day staleness check.
- Cut retained source documents to zero by encrypting extracted text with Fernet before Firestore writes and discarding uploaded PDF bytes.
- Built idempotent ingestion across 8 campus data sources – 880 menu items over 6 dining halls, 1,100+ student organizations, 5,800+ course listings – using deterministic node IDs, conflict-tolerant loaders, and per-record savepoints.
- Kept time-sensitive data current across refresh cadences from 10 minutes to weekly, using scheduled jobs, freshness tests, and serve-stale fallbacks that keep answers available when an upstream source is down.
- Built an LLM tool-calling layer over deterministic campus retrieval, routing natural-language queries to structured tools and verifying generated answers against retrieved context before returning them.

Projects

Agent Governance Runtime – TypeScript, Hono, SQLite, Node.js

2026

- Built a policy-enforcement runtime gating agent tool calls across 18 REST routes and a 12-table SQLite schema, by isolating the decision core into 7 acyclic modules with zero LLM or network imports, enforced by tests that scan source for forbidden module specifiers.
- Built a tamper-evident audit log and offline verifier that re-derives every historical policy decision without trusting the runtime, validated by 15 adversarial tests that forge records a naive verifier would accept.
- Hardened the approval gateway against concurrent-writer races via **BEGIN IMMEDIATE** and compare-and-swap transitions, proven by tests where 40 simultaneous appends yield one contiguous chain and two racing approvers yield exactly one execution.
- Eliminated untraceable side effects across process crashes by committing execution state before backend dispatch and adding a boot-time pass that fails abandoned attempts, under 217 test cases spanning failure and boundary paths.

AI YouTube Tutor – TypeScript, Chrome Extension APIs (MV3), Express

2026

- Fixed a cross-video race where one video’s transcript could answer another’s question, by re-validating tab URL, video, and session ID at all five await boundaries and treating extension storage, not MV3 worker memory, as the source of truth.
- Serialized per-tab state mutations in a promise-tail queue so reads, writes, and broadcasts cannot interleave within a tab while other tabs run in parallel, routing in-flight requests through a registry with cancellation and duplicate rejection.

Physical AI and Robot Learning – Python, LeRobot, Isaac Sim, Orbbec SDK, Docker

2026

- Assembled and calibrated SO-101 leader/follower arms and an Orbbec Astra 2 Pro RGB-D camera: motor-ID assignment, serial communication, controller configuration, teleoperation testing, and depth-recording validation.
- Deployed Isaac Sim 6.0.1 on a remote DGX Spark via Docker and browser-based streaming, configuring environments with and without domain randomization for sim-to-real work.

Leadership

Co-Founder, Badger Cricket Club – UW–Madison (2026). Coordinate practices, logistics, and member communication.

Business Development Associate – Settelo Realty, India (2024–2025). Managed digital campaigns and lead generation.